Unit 2 Guided Notes — Exponents & Scientific Notation

Name _____ Date _____ Period _____

Same base? Move the exponents. Big or tiny number? Pack it into a $\times 10^n$.

🎯 By the end of this unit, I can…

- Apply the product, quotient, and power rules of exponents
- Simplify expressions with negative and zero exponents
- Write and compute with numbers in scientific notation

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 Exponent Rules

📖 Words to know

Base: the number being multiplied. In 2^5, the base is 2.

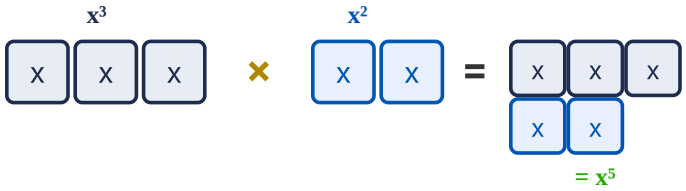
Exponent: the small number up high. It tells how many times to multiply the base. In 2^5, the exponent is 5.

Coefficient: the number in front of a variable. In $4x^3$, the coefficient is 4.

Monomial: one term: a number, a variable, or a product of them, like $4x^3$ or $-2xy^2$.

📌 **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Product Rule — Count the Factors



$x^3 \times x^2 = x^{3+2} = x^5$

Count the factors: 3 + 2 = 5 → ADD the exponents

The Five Core Rules

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	_____	_____	_____
Quotient Rule	$x^a \div x^b = x^{a-b}$	_____	_____	_____
Power Rule	$(x^a)^b = x^{a \cdot b}$ (multiply the exponents)	_____	_____	_____
Zero Exponent	$x^0 = 1$	_____	_____	_____
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	_____	_____	_____

⚠️ **PARENTHESES MATTER!** -3^2 = _____ vs. $(-3)^2$ = _____. Without ( ), ONLY the base gets the exponent — not the negative sign!

Modeling the Rules — See WHY

Each rule comes from writing out the repeated multiplication.

Product Rule · $x^3 \times x^2 = x^5$

Quotient Rule · $x^5 \div x^2 = x^3$

Power Rule · $(x^2)^3 = x^6$

Zero Exponent · $x^0 = 1$

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

- ○ x^8 (Product Rule)
- ○ x^{15} (Power Rule)
- ○ x^2 (Quotient Rule)

Match the Rule to its Key Phrase

MATCH Write the letter of each rule's key phrase on the line.

_____ 1. Product Rule	**A.** Power to power? MULTIPLY
_____ 2. Quotient Rule	**B.** Same base? ADD exponents
_____ 3. Power Rule	**C.** Flip to a fraction
_____ 4. Negative Exponent	**D.** Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

ERROR ANALYSIS Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?

2 Scientific Notation

📖 **Words to know**

Standard form: a number written out with all its digits, like 45,000,000.

Scientific notation: a number written as (a number from 1 up to 10) × 10 to a power, like 4.5×10^7.

📌 **THE FORMAT:** A number in scientific notation looks like 4.5×10^7. The first number must be at least 1 and less than 10. The base is always 10.

Move the Decimal

45,000 (big number) — move LEFT 4 → 4.5×10^4 (exponent +4)

0.0067 (small number) — move RIGHT 3 → 6.7×10^{-3} (exponent −3)

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45{,}000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

Multiplying — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: 3 × 2 = 6

ADD exponents: 4 + 5 = 9

✓ **Answer:** 6×10^{9}

Dividing — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^{6}) \div (4 \times 10^{2})$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^{4}

⚠️ **ADJUSTMENT RULES:** If coefficient ≥ 10: move decimal LEFT, ADD 1 to exponent. If coefficient < 1: move decimal RIGHT, SUBTRACT 1 from exponent.

Example needing adjustment: $(5 \times 10^{3}) \times (4 \times 10^{2})$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^{5}

Adjust (20 ≥ 10): $20 = 2.0 \times 10^{1}$, so ADD 1 to the exponent

✓ **Final answer:** 2.0×10^{6}

ERROR ANALYSIS Convert 0.0045 to scientific notation. A student wrote 4.5×10^{3}. What's the mistake?

✏ Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Exponent Rules with Monomials

1) Simplify $x^3 \cdot x^5$

Answer: _____

2) Simplify $(y^4)^3$

Answer: _____

3) Simplify $a^8 \div a^3$

Answer: _____

4) Simplify $(2x^3)(5x^2)$

Answer: _____

5) Simplify $(3m^2)^3$

Answer: _____

6) Simplify $12y^7 \div 4y^2$

Answer: _____

7) Simplify $(-2a^4)^2$

Answer: _____

8) Simplify $(5x^2y)(3xy^3)$

Answer: _____

9) Simplify $18n^6 \div 6n^6$

Answer: _____

10) Simplify $(4b^3)^2 \div 2b$

Answer: _____

11) Simplify $(-3x^2)(-4x^3)$

Answer: _____

12) Simplify $(x^5y^2)^3$

Answer: _____

B. Scientific Notation Conversions

1) 45,000,000 → scientific notation

Answer: _____

2) 0.00032 → scientific notation

Answer: _____

3) 6.02×10^8 → standard form

Answer: _____

4) 9.1×10^{-5} → standard form

Answer: _____

5) 780,000 → scientific notation

Answer: _____

6) 0.0000067 → scientific notation

Answer: _____

7) 3.45×10^{4} → standard form

Answer: _____

8) 1.2×10^{-3} → standard form

Answer: _____

9) 5,280,000,000 → scientific notation

Answer: _____

10) 2.5×10^{-2} → standard form

Answer: _____

C. Operations

1) $(3 \times 10^{4})(2 \times 10^{5})$

Answer: _____

2) $(8 \times 10^{7}) \div (4 \times 10^{3})$

Answer: _____

3) $(5 \times 10^3)(7 \times 10^6)$

Answer: _____

4) $(9.6 \times 10^8) \div (3.2 \times 10^5)$

Answer: _____

D. Word Problems

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?

2. A bacterium is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacterium?

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^8 people. What is the debt per person?

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

6. A computer does 4.8×10^{9} calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?